

TUESDAY 10/14/25 & THURSDAY 10/16/25
WRITE SIMPLE (NO REGISTERS)

WRITE 0x12 TO DEVICE 0x34

CLOCK IS ALWAYS
SENT BY THE MASTER

BOTH MASTER & DEVICE
CAN SEND DATA &
ACKNOWLEDGES

MASTER

START CONDITION: UNIQUE
CLOCK / DATA COMBINATION

DEVICE

ADD7: 7-BIT ADDRESS
(MSB FIRST + R/W BIT)
0 = WRITE 1 = READ

0110100

< START > < ADD7 > < R/W >

LETS DEVICE KNOW
THE DIRECTION OF FLOW
OF THE DATA

8-BIT DATA
TRANSMISSION

< DATA 8 >

< STOP >

ACK IS ACTIVE-LOW BECAUSE
PULL UP RESISTOR MAKES IT
A "1" IF DEVICE DOES
NOTHING

< ACK >
0

< ACK >
0

REPEATED 1 TO N TIMES

BEFORE START CONDITION IS SET, ASSUME THESE
ARE ALREADY WRITTEN (FOR ALL CASES)

ADD7: ADDRESS NUMBER WRITTEN TO REGISTER 0

REG: REGISTER NUMBER SET IF USING REGISTER
(USE_REG = 1)

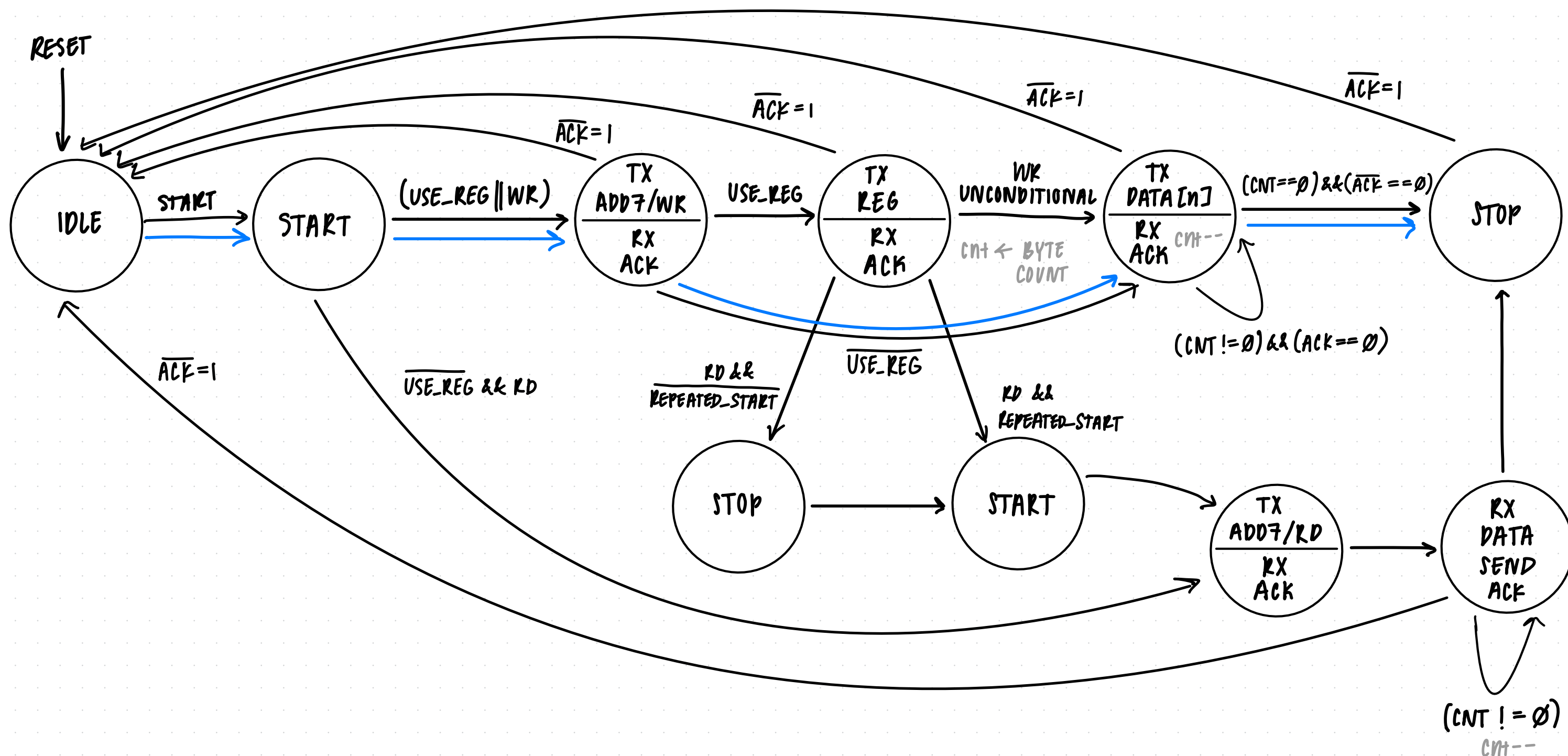
DATA[N]: ALL DATA BEING TRANSMITTED IF
WRITING (R/W = 0)

BYTE_COUNT: NUMBER OF BYTES TO TRANSMIT

USE_REG: FOR READ & WRITE COMPLEX

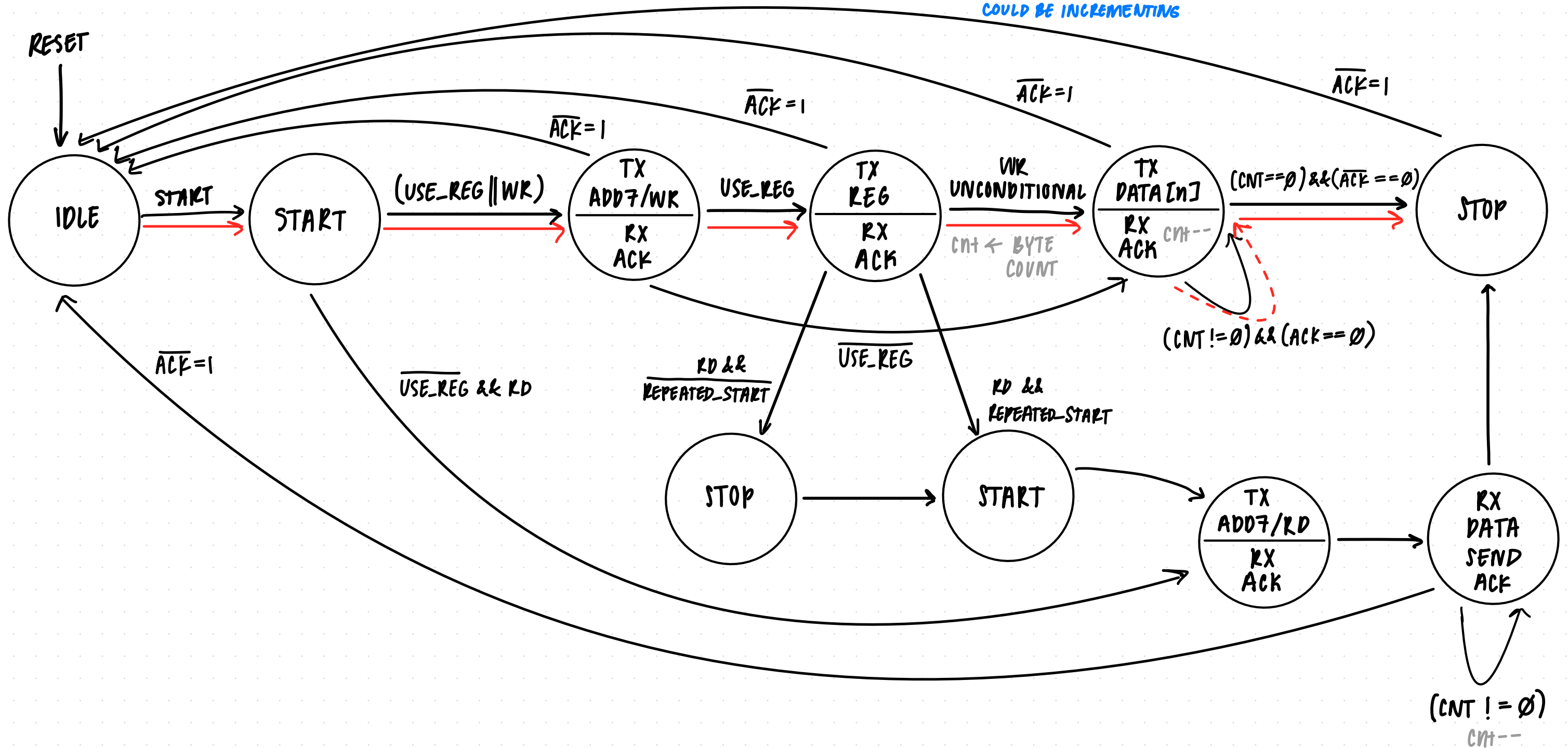
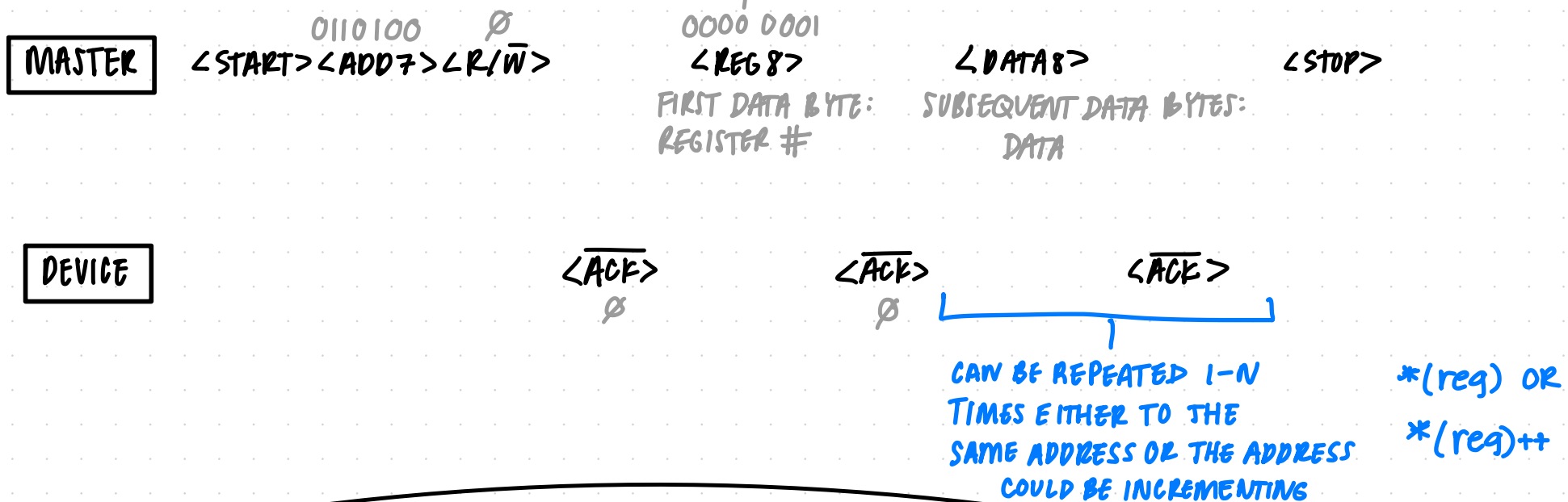
USE_REPEATED_START: FOR READ ONLY

R/W: READ = 1, WRITE = 0



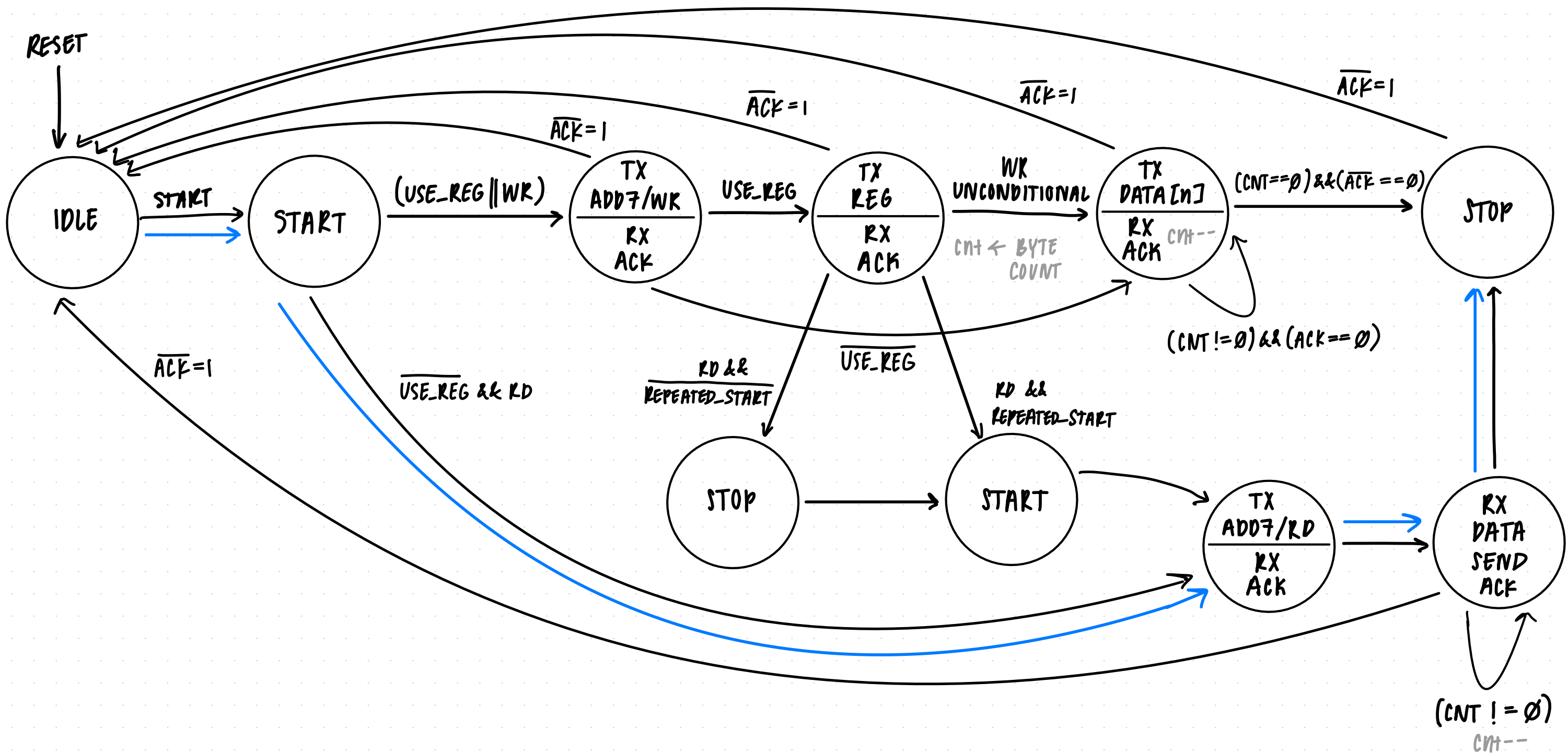
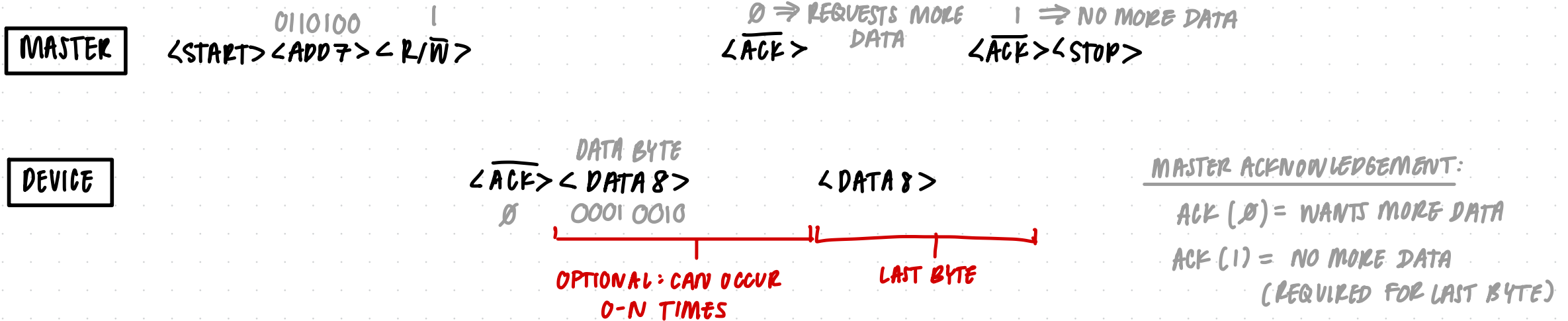
WRITE COMPLEX (WITH REGISTER)

WRITE 0x12 TO DEVICE 0x34



READ SIMPLE (NO REGISTERS)

READ BYTE 0x12 FROM DEVICE 0x34



READ COMPLEX (WITH REGISTER)

DEVICE 0x34
ADDRESS 1
DATA 0x12

MASTER

DEVICE

0110100 0
<START> <ADD7> <R/W>
WRITE THE REGISTER NUMBER FROM WHICH DATA MUST BE READ

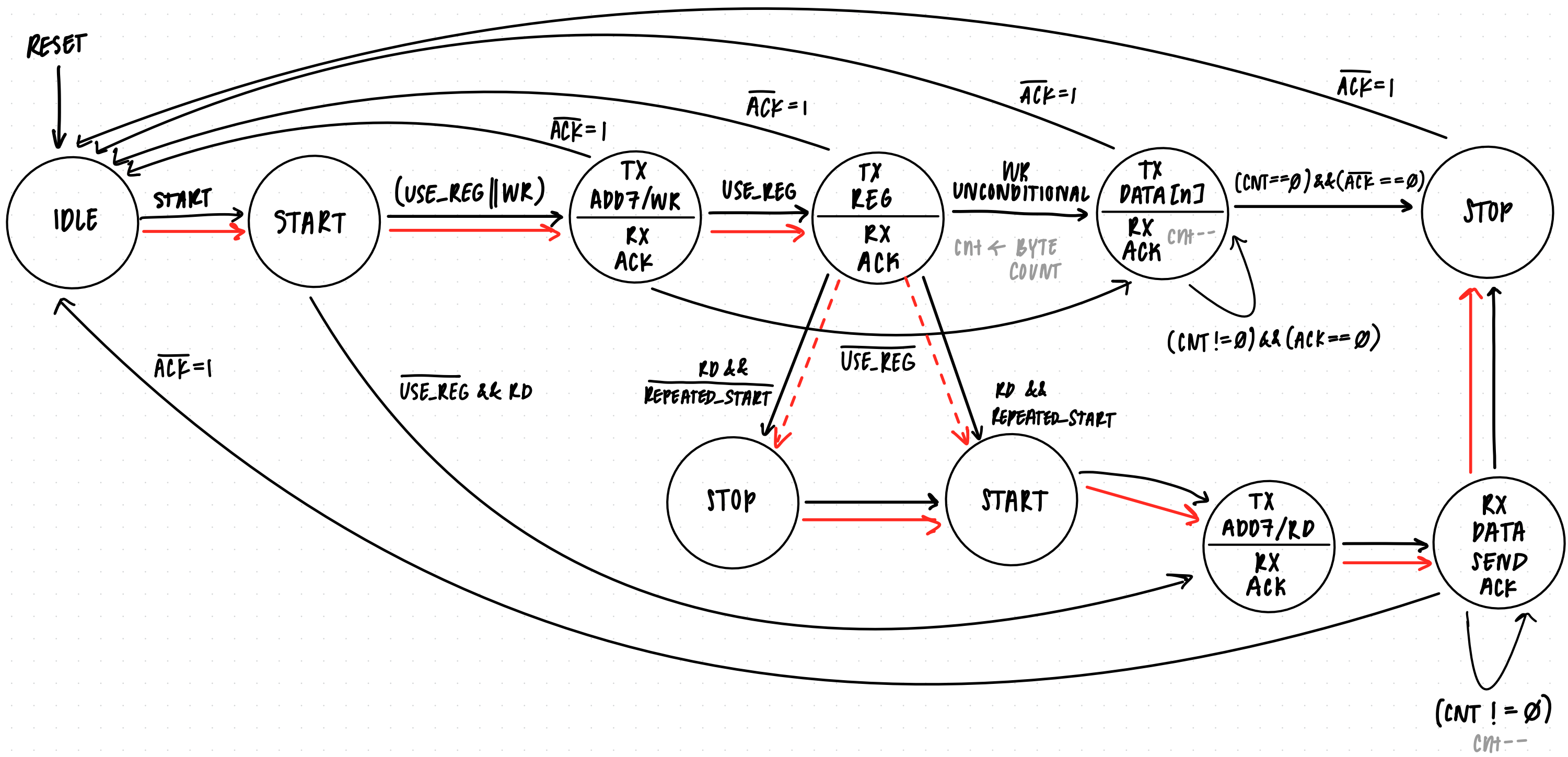
0000 0001
<REG8>
OPTIONAL DEPENDING ON DEVICE
REVERSE BACK TO READING AGAIN

1
<STOP> <START> <ADD7> <R/W>
REPEATED START: STARTING WITHOUT A STOP CONDITION BEFORE IT

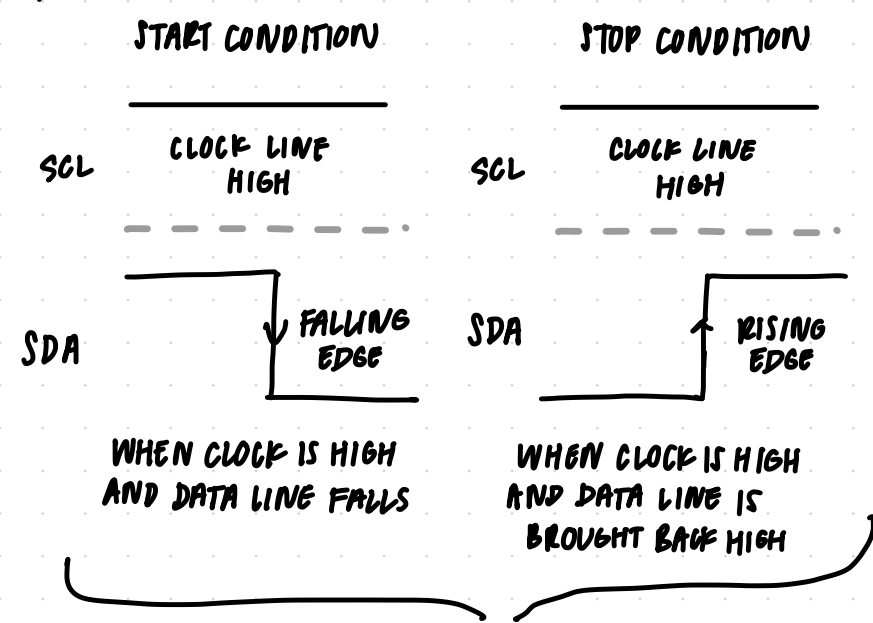
0
<ACK>
MUST SEND NEGATIVE ACKNOWLEDGEMENT
CANNOT GO FROM ACK 0 TO STOP

1 ⇒ NO MORE DATA
<ACK> <STOP>

<ACK> 0
<ACK> 0
<ACK> <DATA8> 0 00010010 (0x12)
<DATA8>
CAN BE SENT 0-N TIMES



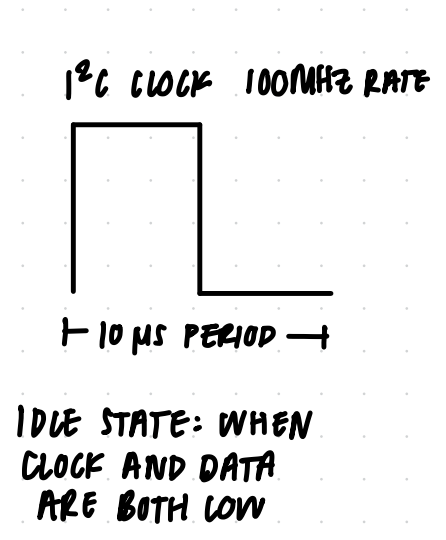
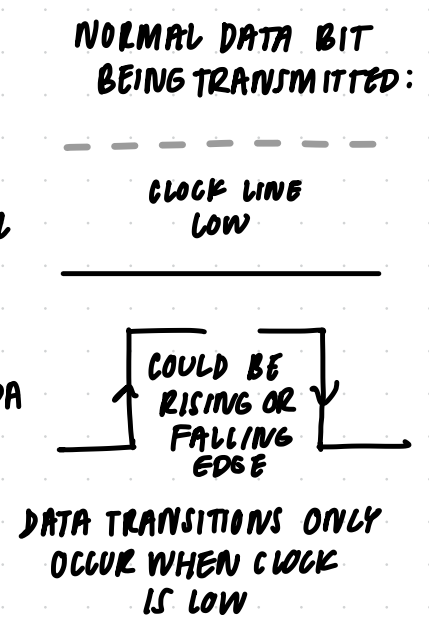
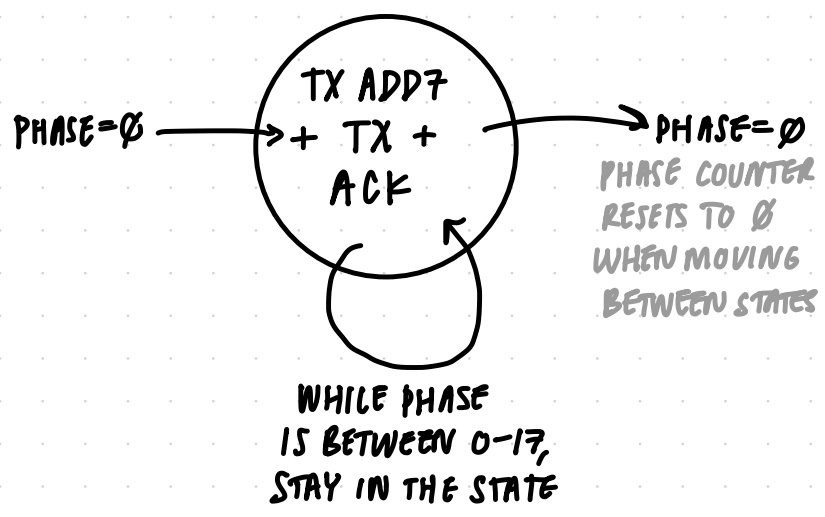
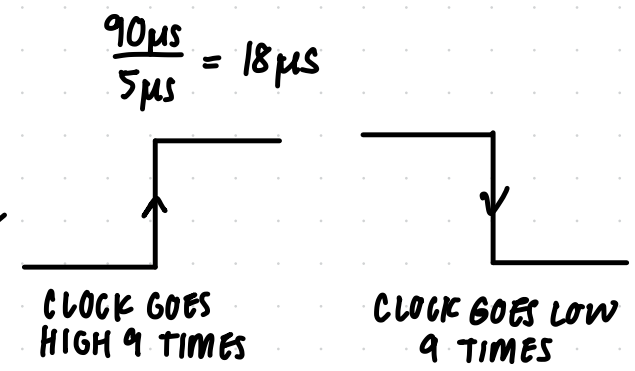
TUESDAY 10/21/25



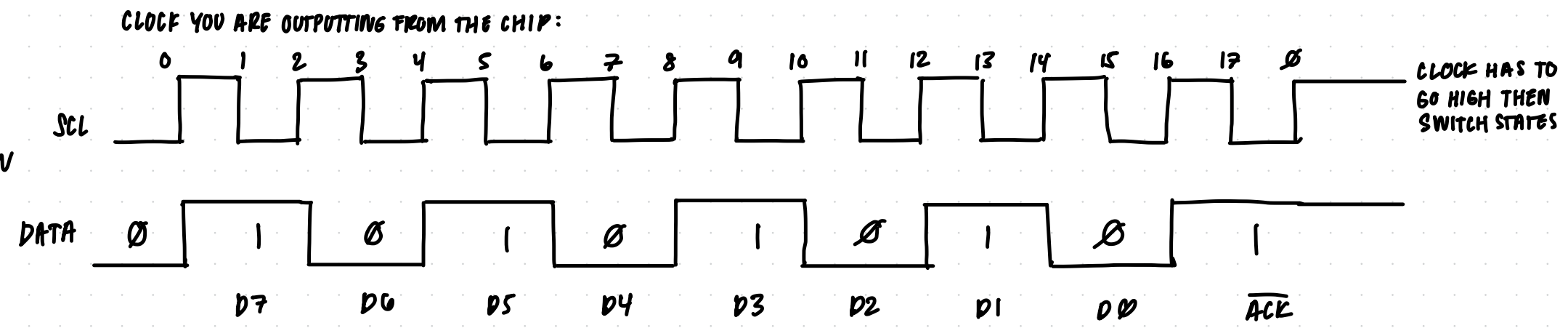
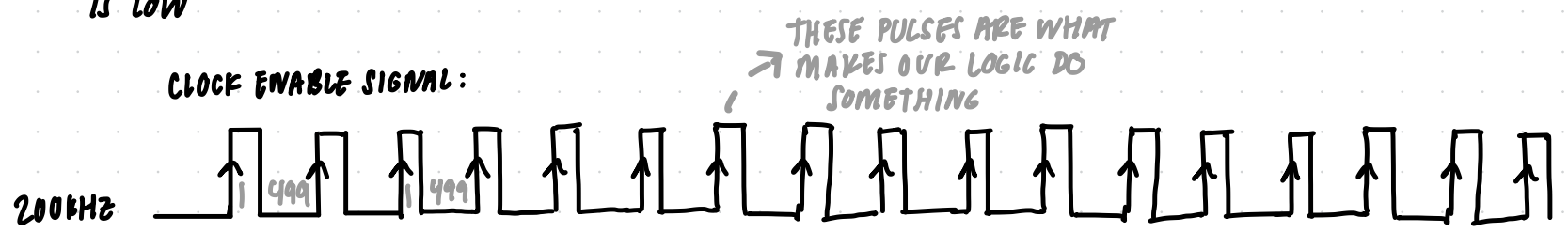
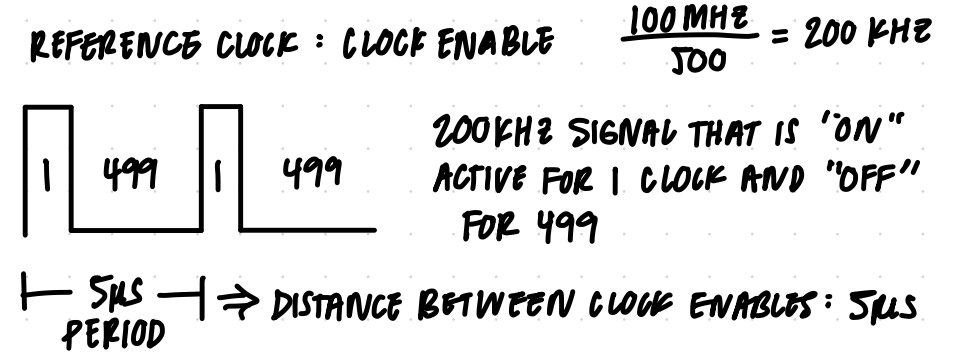
YOU GENERALLY NEVER HAVE A DATA LINE TRANSITION WHEN CLOCK IS HIGH. IT ONLY HAPPENS FOR THESE TWO CONDITIONS (START & STOP)

18 CLOCKS OCCUR IN EACH STATE:

(8 BITS BEING TRANSMITTED + 1 BIT BEING RECEIVED) x 10μS = 90μS PER TRANSMISSION STATE



IDLE STATE: WHEN CLOCK AND DATA ARE BOTH LOW



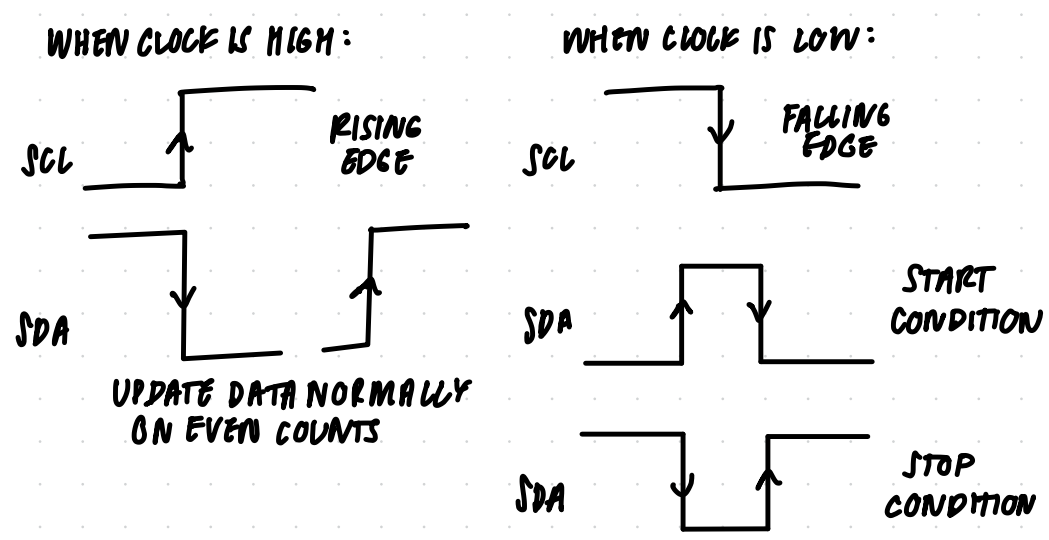
PHASE: UNDERLYING COUNTER THAT KEEPS TRACK OF HOW FAR YOU ARE IN EACH STATE (HOW MANY 5μS PERIODS THAT HAVE ELAPSED)

IDENTICAL TIMING FOR TRANSMIT AND RECEIVE: ONLY DATA DIRECTION DIFFERS

READ & UPDATE TRANSMITTER VALUE ON EVEN COUNTS

SAMPLE THE SIGNAL ON THE ODD COUNTS

DIFFERENT TIMING ONLY FOR START & STOP CONDITIONS (3 CLOCK CYCLES)



HOW TO MOVE TO NEXT STATE

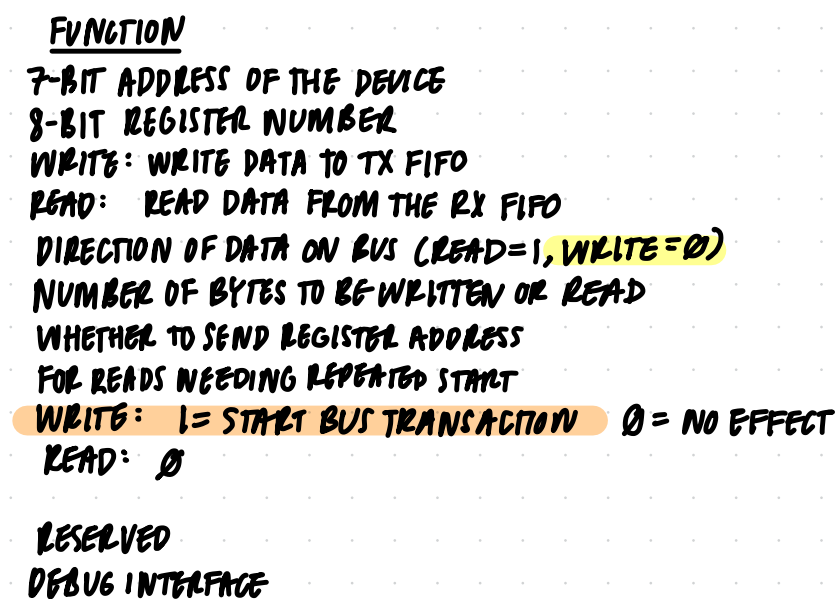
FOR MOST THINGS: COUNT 0-17

IF (COUNT == 17) MOVE ONTO NEXT STATE AND MAKE PHASE=0

FOR START/STOP: COUNT 0-2

IF (COUNT == 2) MAKE PHASE=0 ON NEXT CLOCK THEN MOVE TO NEXT STATE

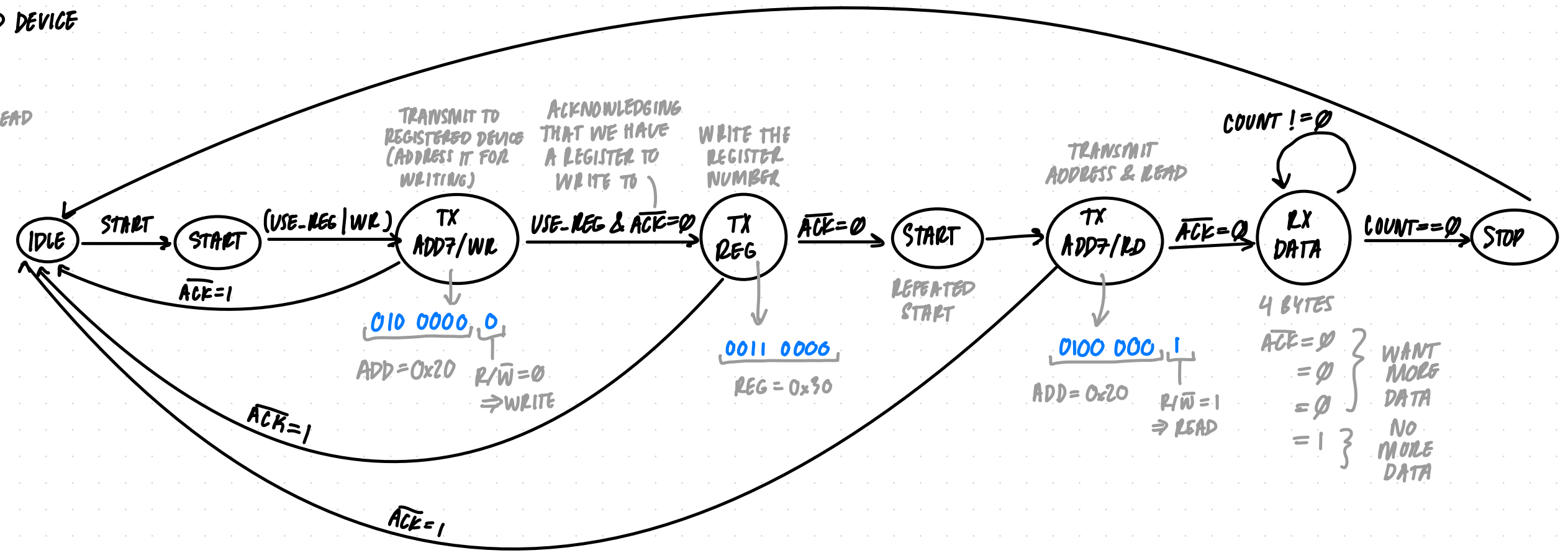
MCP23008
MCP23S08 → SPI VERSION: DO NOT USE



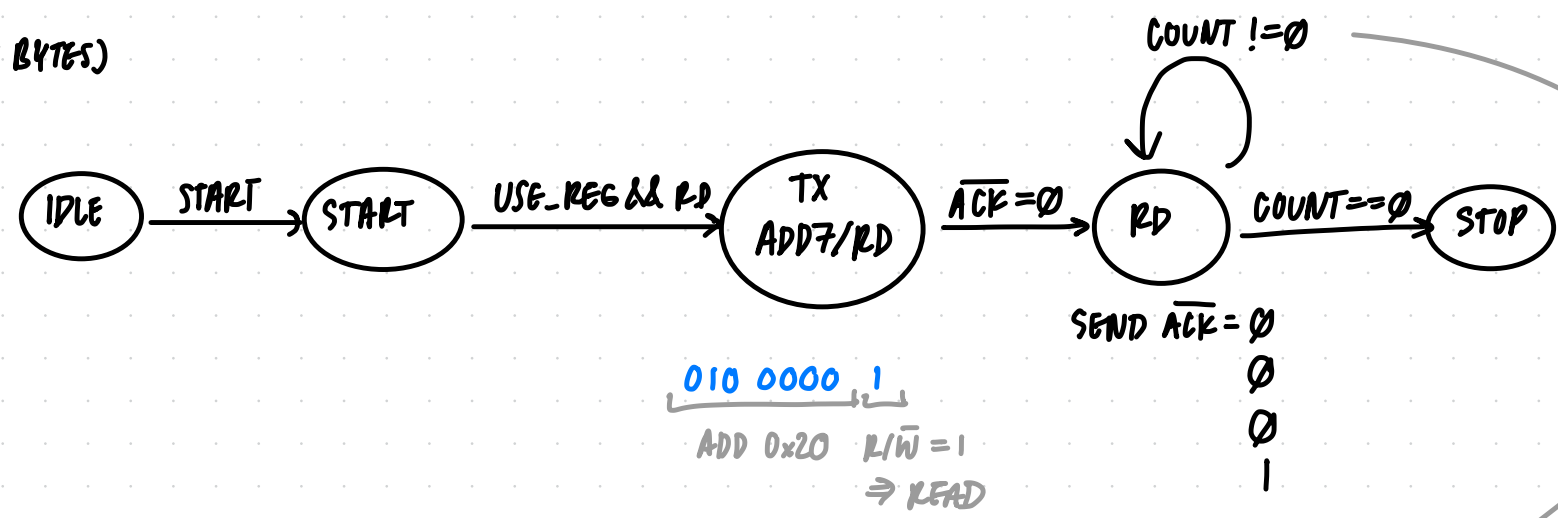
READ 4 BYTES FROM A REGISTERED DEVICE

ADDRESS 0x20
 REPEATED START
 REGISTER 0x30
 0x31
 0x32
 0x33

REGISTERS TO READ
 VALUE FROM

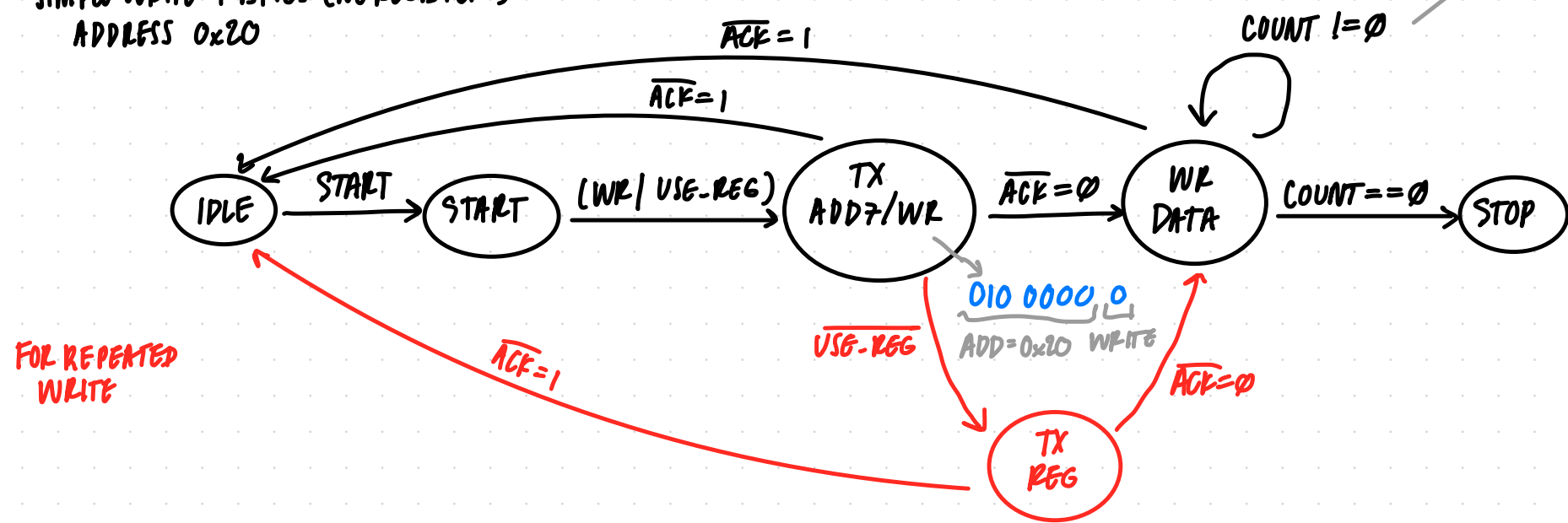


SIMPLE READ (4 BYTES)



COUNT != 0 : REMAINING BYTES TO TRANSMIT / RECEIVE

SIMPLE WRITE 4 BYTES (NO REGISTERS) ADDRESS 0x20



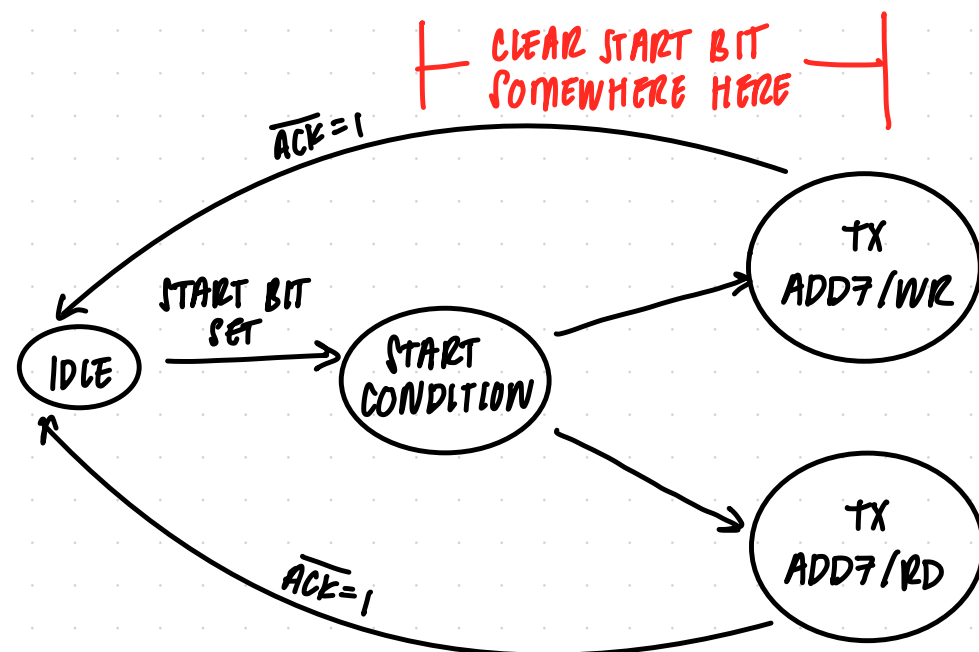
FOR REPEATED WRITE

IMPLEMENTING THE CONTROL REGISTER

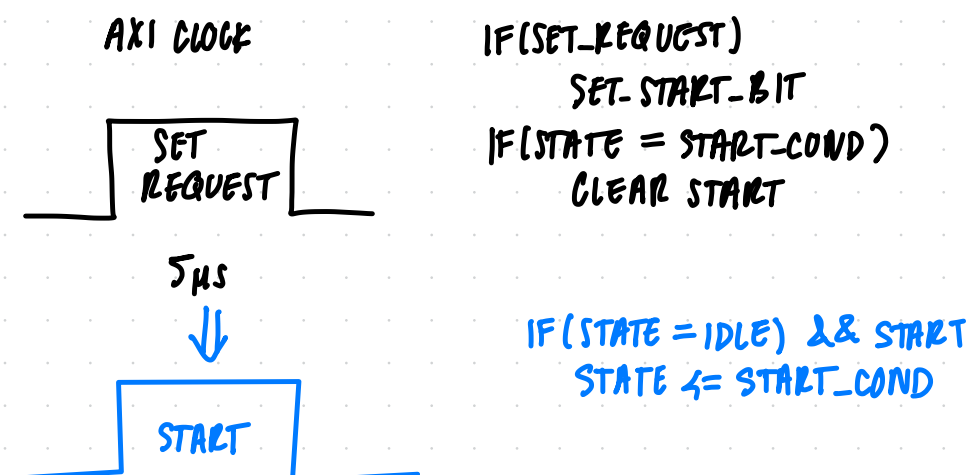
BIT 7: START BIT

- WHEN THE START BIT IS WRITTEN HIGH, THE FSM EXITS IDLE AND BEGINS THE START CONDITION. IF THE START BIT REMAINS HIGH WHEN THE FSM RETURNS TO IDLE, IT WILL IMMEDIATELY RE-ENTER THE START CONDITION, CAUSING REPEATED RETRANSMISSIONS
- TURNED ON WHEN YOU WRITE TO THE REGISTER AND AUTO-CLEARING ITSELF IN THE STATE MACHINE BEFORE YOU GET BACK TO IDLE
- CLEAR NO LATER THAN TX ADD/WR OR TX ADD/RD

MUST CLEAR START BIT BETWEEN START CONDITION AND ADDRESS TRANSMISSION STATES TO PREVENT INFINITE RETRANSMISSION LOOPS



ONLY 1 PROCEDURAL BLOCK CAN BE USED TO MODIFY THE START BIT
START CONDITION IS MANAGED BY THE FSM



EXTEND THE START PULSE TO LAST AT LEAST ONE FULL PERIOD OF THE SLOWER FSM CLOCK ENABLE TO GUARANTEE ITS DETECTED

METHOD 1: PULSE EXTENSION

- AXI CLOCK (100 MHz) SETS START BIT FOR EXACTLY 500 CLOCKS (5 μ s)
- FSM RUNNING AT 200 KHz IS GUARANTEED TO SEE THE PULSE
- AUTOMATICALLY CLEARS AFTER TIMEOUT

ON POSEDGE AXI CLK

IF(RESET)

IF(WR && ADD=CONTROL_REG)

SET_START_REQUEST <= 1

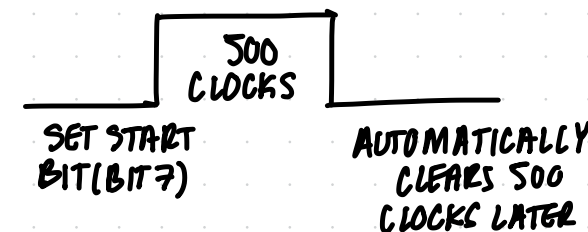
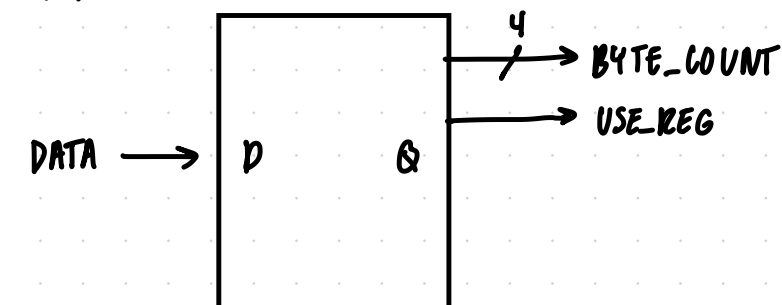
ELSE IF(COUNT==500)

SET_START_REQUEST <= 0

CREATES AN AUTO RESETTING REQUEST EVERY 10ns

HOWEVER, SINCE THE FSM RUNS AT 200 KHz, IT MAY NOT DETECT SUCH A SHORT PULSE, LEADING TO MISSED REQUESTS

FOR OTHER BITS:



METHOD 2: SEND CLEAR START REQUEST FROM SLOWER CLOCK TO THE FAST CLOCK

AXI CLOCK MANAGES START BIT SETTING & CLEARING

AXI CLK RUNNING AT 100 MHz

IF(RESET)

PRIORITY 1: RESET

ELSE IF(WR && ADD=CTRL) -> PRIORITY 2: WRITING TO THE BUS

SET START BIT

ELSE IF(CLR_START_REQ) -> PRIORITY 3: CLEAR REQUEST

CLEAR START BIT

START <= 0

FSM SENDS CLEAR REQUEST BACK TO AXI CLOCK WHEN IN START CONDITION

FSM @ 200 KHz

IF(STATE==START_COND)

CLEAR_START_REQ <= 1

ELSE

CLEAR_START_REQ <= 0

WHEN CLEAR REQUEST OCCURS IT IS ACTIVE FOR 500 CLOCKS SO IF YOU PUT IT HIGHER THAN THE WR PROCESSING, YOU IGNORE WRITES/READS TO THE AXI BUS FOR 500 CLOCKS

WHILE CLEAR REQUEST IS ACTIVE, AXI READ/WRITE ACCESSSES TO THE CONTROL REGISTER ARE IGNORED TO AVOID CONFLICTS WITH FSM

SDA PIN STATES:

$\bar{Z}$: HIGH IMPEDANCE

0 :

NEVER TRANSMIT A "1"

↳ IF 1 & 0 ARE DRIVEN : SHORT CIRCUIT

↳ IF $\bar{Z}$ & 0 ARE DRIVEN : 0 WINS OVER PULL UP

DATA STATES & SENDING DATA

$PIN \leftarrow DATA_RX[i]$ ← SHOULD BE A FUNCTION OF THE PHASE

IF (DATA_RX[i])

$PIN \leftarrow \bar{Z}$

ELSE

$PIN \leftarrow 0$

VERILOG ALLOWS A PIN TO HAVE 4 STATES:

0

1

$\bar{Z}$: HIGH IMPEDANCE

X : DONT CARE

↳ USE WHATEVER REQUIRES LESS GATES

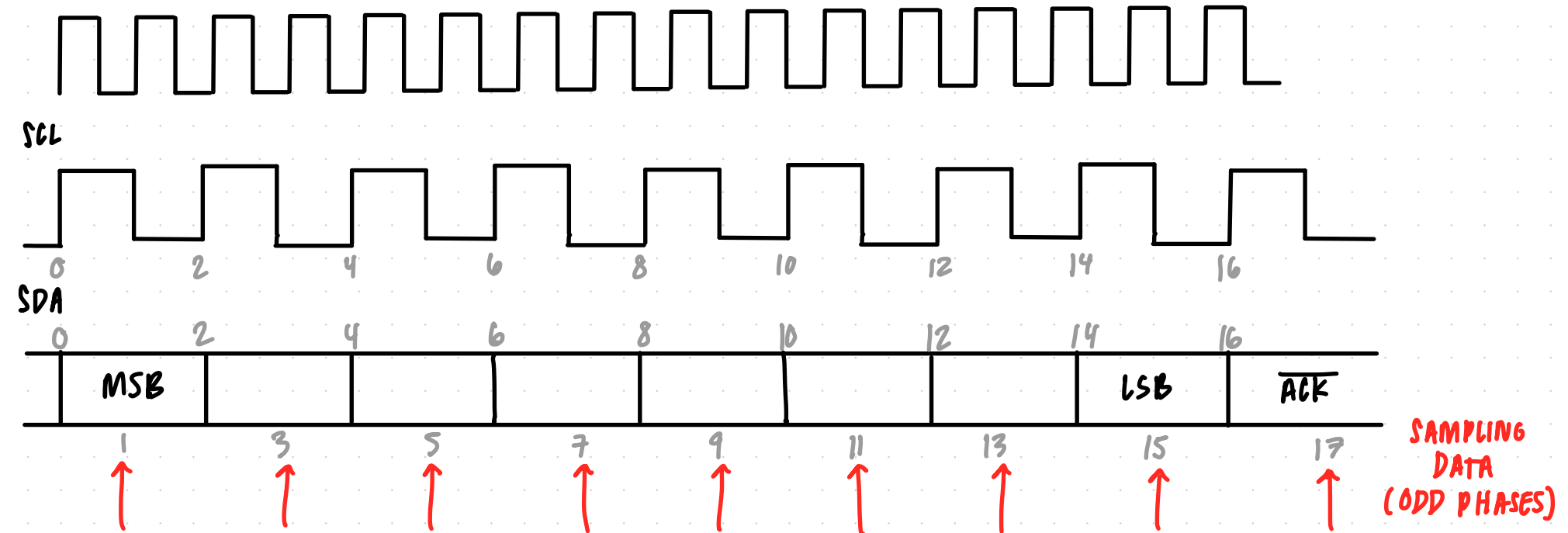
DATA TRANSMISSION TIMING

DATA UPDATES: EVEN PHASES (0, 2, 4, 6, 8, 10, 12, 14) RISING EDGE OF CLOCK

DATA SAMPLING: ODD PHASES (1, 3, 5, 7, 9, 11, 13, 15)

ACK BIT: UPDATED AT PHASE 16, READ AT PHASE 17 (FALLING EDGE OF AXI CLOCK)

200 KHz CLOCK ENABLE



FIFO INTEGRATION

EITHER LATCH THE DATA AND READ REQUEST TO THAT FIFO TO ADVANCE THE POINTER FORWARD

OR

DONT DO A READ REQUEST UNTIL CLOCK 16

(BECAUSE ALL DATA HAS ALREADY BEEN SENT AND THERE IS NO USE FOR WHAT IS CURRENTLY BEING OUTPUTTED)

AFTER RECEIVING A FULL BYTE (8 BITS), LATCH THE RECEIVED DATA INTO A REGISTER

DURING OR IMMEDIATELY AFTER THE ACK CYCLE, PULSE FIFO WRITE ENABLE

PULSING THE WRITE REQUEST DURING THE ACK CYCLE ENSURES YOU CAPTURE THE BYTE AFTER IT'S COMPLETE AND BEFORE THE NEXT BYTE BEGINS